GENERAL DESCRIPTION

A NEW FLOW THROUGH IRRIGATION POND AND IRRIGATION PUMP STATION IS TO PROVIDE WATER TO THE EXISTING AND FUTURE IRRIGATION SYSTEM FOR NEWTON–ATHENS. THE FOLLOWING CONTRACTOR RESPONSIBILITIES INCLUDE:

1. CONNECT NEW SUPPLY PIPE TO EXISTING RAW WATER VAULT TO CONVEY WATER TO NEW IRRIGATION POND. THE EXISTING CONCRETE PIPE DOWNSTREAM OF THE EXISTING RAW WATER PIPE IS PLUGGED AND ABANDONED IN PLACE. ALL WATER IS TO FLOW FROM THE VAULT TO THE NEW POND THEN THROUGH AN OVERFLOW PIPE THAT RECONNECTS TO EXISTING DOWNSTREAM VAULT. REFER TO PLANS.
2. CONSTRUCT NEW LINED IRRIGATION POND.
3. INSTALLATION OF INTAKE STRUCTURE, INTAKE PIPE, WET WELL, AND NEW IRRIGATION PUMP STATION.
4. CONNECT NEW MAINLINE TO EXISTING MAINLINES AND STUB OUT NEW MAINLINE FOR FUTURE IRRIGATION SYSTEM.
5. DEMOLITION OF EXISTING PUMP STATION.
6. OPTIONAL EQUIPMENT: AERATION SYSTEM FOR NEW POND. OWNER TO DETERMINE IF THIS ITEM IS DESIRED.

GENERAL NOTES

1. THE PUMP STATION FLOW, PRESSURE, AND VOLTAGE REQUIREMENTS ARE INDICATED IN THE LEGEND.
2. THE PUMP STATION SKID CONTAINS TWO MAIN PUMPS AND ONE PRESSURE MAINTENANCE PUMP.
3. VERIFY EXISTING SITE CONDITIONS PRIOR TO CONSTRUCTION AND BRING ANY CONCERNS TO THE OWNER'S REPRESENTATIVE.
4. READ THOROUGHLY AND BECOME FAMILIAR WITH THE SPECIFICATIONS AND INSTALLATION DETAILS FOR THIS AND RELATED WORK PRIOR TO CONSTRUCTION.
5. COORDINATE UTILITY LOCATES ("CALL BEFORE YOU DIG") OF UNDERGROUND UTILITIES PRIOR TO CONSTRUCTION.
6. DO NOT PROCEED WITH THE INSTALLATION OF THE SYSTEM WHEN IT IS OBVIOUS IN THE FIELD THAT OBSTRUCTIONS OR GRADE DIFFERENCES EXIST THAT MIGHT NOT HAVE BEEN CONSIDERED IN THE ENGINEERING, OR IF DISCREPANCIES IN CONSTRUCTION DETAILS, LEGEND, NOTES, OR SPECIFICATIONS ARE DISCOVERED. BRING ALL SUCH OBSTRUCTIONS OR DISCREPANCIES TO THE ATTENTION OF THE OWNER'S REPRESENTATIVE.
7. THE DRAWINGS ARE DIAGRAMMATIC. SYSTEM COMPONENTS MAY BE SHOWN OUTSIDE PLANTING AREAS FOR GRAPHIC CLARITY. AVOID CONFLICTS BETWEEN THE SYSTEM, PLANTING MATERIALS, AND ARCHITECTURAL FEATURES. INSTALL SYSTEM COMPONENTS IN LANDSCAPED AREAS WHEREVER POSSIBLE.
8. INSTALL ELECTRICAL POWER TO THE PUMP SYSTEM IN ACCORDANCE WITH NEC AND OTHER APPLICABLE LOCAL ELECTRIC UTILITY CODES.
9. WITH REGARD TO PIPE SIZING, THE FOLLOWING SHOULD BE NOTED:
   IF A SECTION OF UNSIZED PIPE IS LOCATED BETWEEN TWO IDENTICALLY SIZED SECTIONS, THE UNSIZED PIPE IS THE SAME NOMINAL SIZE AS THE TWO SIZED SECTIONS AND SHOULD NOT BE CONFUSED WITH THE DEFAULT PIPE SIZE NOTED IN THE LEGEND.
10. THE OPTIONAL POND AERATION SYSTEM IS INTENDED TO OPERATE 12 HOURS PER DAY AND PROVIDE FOUR (4) TURNOVERS PER DAY.

LEGEND

- Ⓥ — EXISTING CONCRETE VAULT
  *APPROXIMATE LOCATION
- – – – – — EXISTING RAW WATER DELIVERY PIPE
  *12–INCH CONCRETE PIPE
  *APPROXIMATE LOCATION
  *0.3 CFS (134 GPM) CONTINUOUS FLOW TYPICAL
- SUPPLY PIPE
  *CLASS 200 PVC
  *SIZE: PER PLAN
- INTAKE PIPE
  *C905 PVC
  *SIZE: PER PLAN
- OVERFLOW PIPE
  *CLASS 200 PVC
  *SIZE: PER PLAN
- MAINLINE PIPE
  *CLASS 200 PVC
  *SIZE: PER PLAN
- — B — BACKWASH PIPE FOR INTAKE SCREEN
  *1–1/2–INCH 160 PSI POLYETHYLENE PIPE
- — F — FILTER FLUSH PIPE
  *TYPE: CLASS 200 PVC BELOW GRADE, SCH. 40 STEEL ABOVE GRADE
  *SIZE: 4–INCH
- AERATION SLEEVE
  *TYPE: CLASS 200 PVC
  *SIZE: 8–INCH
- [S] INTAKE STRUCTURE
  *LAKOS PLUM CREEK PC3424
  *30 MESH SCREEN
  *RATED AT 1700 GPM
  *16–INCH FLANGED END
- Ⓟ PUMP STATION
  *FLOW: 900 GPM
  *PRESSURE: 100 PSI DISCHARGE
  *460/480 VAC, 3 PHASE
  *2, 40 HP MAINS. 1, 3 HP PRESSURE MAINTENANCE PUMP
  *HOUSED WITHIN DESIGNED STRUCTURE (BY OTHERS)
- Ⓞ OVERFLOW ASSEMBLY
- ⊗ ISOLATION GATE VALVE ASSEMBLY
  *CLOW OR EQUAL
  *SIZE TO MATCH PIPE SIZE
- ⊙ WINTERIZATION ASSEMBLY
- CAPPED PIPE
- PIPE CONNECTION

OPTIONAL EQUIPMENT (NOT SHOWN):
1. AERATION SYSTEM (4 CFM)
2. AERATION TUBING (½" WEIGHTED PVC)
3. AERATION POD (2, 2 CFM PODS)

NOTE: SINCE POND IS FLOW THROUGH WATER QUALITY MAY NOT REQUIRE AERATION SYSTEM. TBD.

NOTE: VALVE BOXES TO BE CARSON WITH PURPLE LID

FLAG NOTES

1. EXCAVATE AND EXPOSE EXISTING CONCRETE VAULT. INSTALL NEW SUPPLY PIPE WITH INVERT ELEVATION INDICATED ON PLAN. SEAL PENETRATION WATER TIGHT WITH HYDRAULIC CEMENT OR EQUAL. COORDINATE CONNECTION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
2. PERMANENTLY PLUG EXISTING 12–INCH CONCRETE PIPE AT THE EXISTING VAULT. FILL PIPE AS NECESSARY WITH CONCRETE OR USE WHATEVER MEANS NECESSARY TO PLUG PIPE. TEST PLUG FOR LEAKS BY FILLING EXISTING VAULT WITH WATER AND COMPLETE STATIC TEST OVER 24–HOUR PERIOD. WATER LEVEL NOT TO DROP DURING TEST. SEAL LEAKS IF TEST FAILS. TEST PRIOR TO INSTALLING NEW PIPING TO INTO VAULT. COORDINATE PLUG AND TESTING WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
3. INSTALL SUPPLY PIPE WITH A GRADUAL SLOPE OF APPROXIMATELY 2.4%. REFER TO INVERTS INDICATED ON PLAN.
4. SEAL PIPE PENETRATION THROUGH LINER WITH BOOT AND STAINLESS STEEL STRAPS PER MANUFACTURER'S GUIDELINES.
5. INSTALL 4–FOOT WIDE X 6–INCH THICK CONCRETE (5000 PSI MIX) APRON FROM SUPPLY PIPE DISCHARGE TO BOTTOM OF POND. EXTEND APRON 10–FEET ALONG BOTTOM OF POND. CENTER APRON IN MIDDLE OF SUPPLY PIPE. INSTALL 6–INCH HIGH X 6–INCH WIDE LIP AROUND PERIMETER OF APRON ROUTED ALONG BOTTOM OF POND INTENDED TO DISSIPATE ENERGY. INTENT OF APRON IS TO MINIMIZE EROSION WHILE FILLING POND.
6. SURVEY POND EXTENTS PRIOR TO POND INSTALLATION. ELEVATIONS SHOWN ON PLAN ARE EXCAVATED ELEVATIONS. INSTALL LINER SYSTEM PER MANUFACTURER'S GUIDELINES. COORDINATE POND INSTALLATION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
7. THE POND CONSTRUCTION WILL REQUIRE EXTENSIVE IRRIGATION DEMOLITION OF EXISTING MAINLINE, WIRING, SPRINKLERS, LATERAL, ETC. EXACT LOCATIONS OF THESE ITEMS ARE NOT KNOWN. CONTRACTOR TO REMOVE ENCOUNTERED ITEMS AND DISPOSE OFF SITE. IF ENCOUNTERED, IDENTIFY PIPE AND SOURCE. ENSURE CUT PIPES DON'T DISCHARGE WATER WHEN PRESSURIZED – PERMANENTLY CAP ENDS. NEW IRRIGATION REQUIRED FOR AREAS NEAR POND. DESIGN TO BE DETERMINED. COORDINATE DEMOLITION ACTIVITIES AND INSTALLATION OF NEW IRRIGATION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
8. INSTALL POND OVERFLOW ASSEMBLY IN APPROXIMATE LOCATION.
9. INSTALL INTAKE STRUCTURE IN APPROXIMATE LOCATION. COORDINATE INSTALLATION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
10. INSTALL ONE (1) AERATION TUBING SLEEVE INTENDED FOR POTENTIAL AERATION SYSTEM. COORDINATE SLEEVE INSTALLATION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
11. INSTALL WET WELL AND PUMP STATION IN APPROXIMATE LOCATION. MAKE ALL ELECTRICAL CONNECTIONS PER NEC CODE AND MANUFACTURER'S GUIDELINES. POWER SOURCE TBD. COORDINATE INSTALLATION OF PUMP STATION WITH OTHER TRADES (ELECTRICAL, ARCHITECTURAL) AND OWNER'S REPRESENTATIVE.
12. FOR FUTURE USE: INSTALL HDPE FLANGE ADAPTER WITH DUCTILE IRON BACKUP RING IN APPROXIMATE LOCATION. SECURE FLANGE TO ADAPTER WITH STAINLESS STEEL BOLTS. PLACE STANDARD RECTANGULAR VALVE BOX OVER CAPPED END. COORDINATE CAP INSTALLATION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
13. EXCAVATE AND EXPOSE EXISTING PVC MAINLINE IN APPROXIMATE LOCATION. EXACT SIZE AND LOCATION NOT KNOWN – FIELD VERIFY. INSTALL NECESSARY FITTINGS TO COMPLETE CONNECTION. COORDINATE CONNECTION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
14. INSTALL SUPPLY PIPE AT A GRADUAL SLOPE OF APPROXIMATELY 1%. REFER TO INVERTS INDICATED ON PLAN.
15. COMPLETELY DEMOLISH EXISTING PUMP STATION. REMOVAL ITEMS INCLUDE BUT ARE NOT LIMITED TO:
    - BUILDING
    - CONTROL PANEL
    - PUMPS
    - MOTORS
    - PIPING
    - DISCONNECT AND REMOVE DOGLEG. PLUG END AS NECESSARY. INSTALL QUICK COUPLING VALVE UPSTREAM OF PLUG TO ASSIST WINTERIZATION OF DEAD END.

    PERMANENTLY PLUG CONCRETE PIPE SUPPLYING WATER (TEE FROM DITCH) THAT SUPPLIES WATER TO VAULT WITH CONCRETE OR EQUAL. HYDROSTATICALLY TEST SO PLUG DOES NOT LEAK. ABANDON VAULT IN PLACE. COORDINATE ITEMS THE OWNER WANTS TO KEEP. DISPOSE OF UNWANTED ITEMS OFF SITE. COORDINATE DEMOLITION WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
16. APPROXIMATELY EIGHT (8) MATURE TREES WILL REQUIRE REMOVAL TO ACCOMMODATE THE NEW POND. CONTRACTOR TO REMOVE AND DISPOSE OF TREES IDENTIFIED WITH OWNER'S REPRESENTATIVE PRIOR TO CONSTRUCTION.
17. PIPE CROSSES EXISTING ASPHALT PARKING LOT. CUT/COLD PATCH PARKING LOT TO INSTALL PIPE. COORDINATE CUT/ COLD PATCH WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.
18. SUPPLY PIPE CROSSES EXISTING TREES SOME OF WHICH MAY BE NEARING END OF LIFE. REMOVE AND DISPOSE OF DYING TREES. IF PIPE CROSSES HEALTH TREES ADJUST ROUTING TO MINIMIZE ROOT IMPACTS. COORDINATE TREE REMOVAL AND PIPE ROUTING WITH THE OWNER'S REPRESENTATIVE ON SITE PRIOR TO CONSTRUCTION.

Aqua Engineering Inc.
Innovative Water Solutions

UNIVERSITY OF COLORADO
NEWTON AND ATHENS RAW WATER
SCHEMATIC DESIGN

RAW WATER COVER SHEET

PRELIMINARY DESIGN NOT FOR CONSTRUCTION

DATE	5-10-2017
DESIGNED BY	JHK
DRAWN BY	JHK
CHECKED BY	CBK

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POINT-OF-CONNECTION
*SOURCE: EXISTING CONCRETE VAULT, RAW WATER FOR IRRIGATION
*FLOW: ±0.3 CFS (134 GPM)
*GRAVITY FED

SUPPLY PIPE
INVERT EL. 5307'

WARNING!! ELECTRICAL CROSSING

OPERATIONAL WATER SURFACE EL. 5304'

PENETRATION INV EL. 5299'

PARK
45,700sf
1.05ac

3
800sf
ac

6
000sf
ac

EXISTING BASKETBALL COURT

EXISTING SAND VOLLEYBALL COURT

EXISTING GARDENS

WARNING!! STORM CROSSING

WARNING!! ELECTRICAL CROSSING

WARNING!! TELECOM CROSSING

WARNING!! ELECTRICAL CROSSING

WARNING!! STORM AND POTABLE CROSSING

CONNECT TO FOOTBALL FIELD IRRIGATION MAINLINE PIPE

APPROXIMATE OVERFLOW PIPE INVERT EL. 5300'

KRUSE–
BOEDECKER
INDOOR
PRACTICE
FACILITY

FOOTBALL PRACTICE FIELDS

30' 0' 15' 30' 60'
1" = 30'
SCALE MUST MEASURE 1 INCH OR DRAWING IS NOT TO SCALE.

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UNIVERSITY OF COLORADO
NEWTON AND ATHENS RAW WATER
SCHEMATIC DESIGN

RAW WATER PLAN

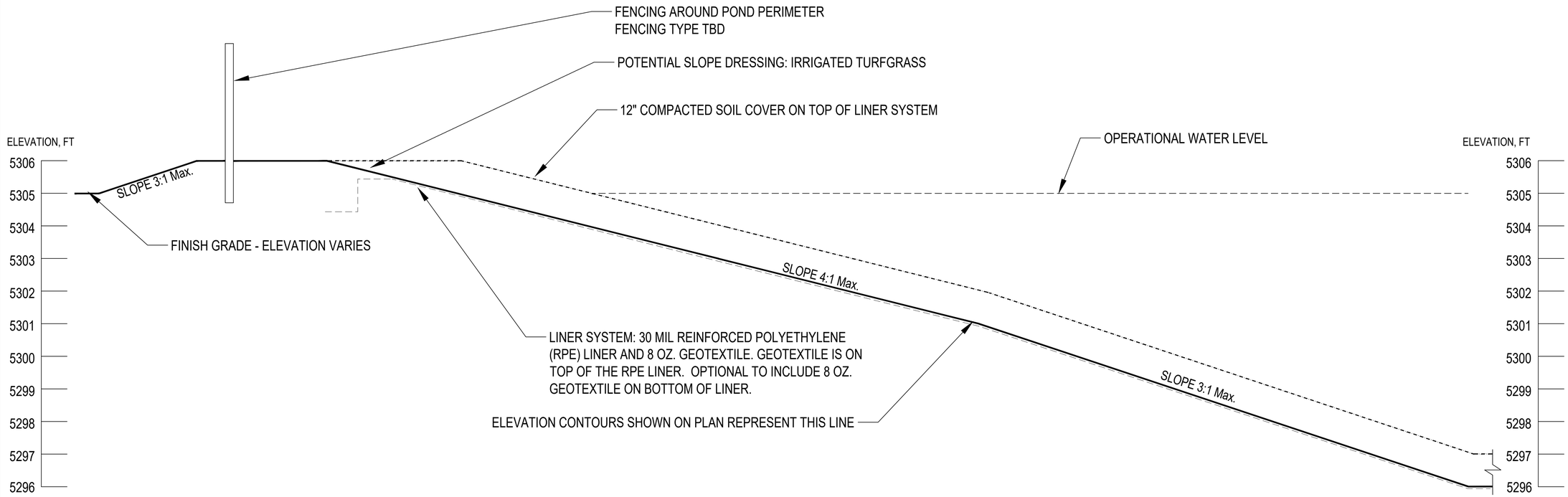
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1 PROPOSED POND PROFILE (EXCAVATED)

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NEWTON AND ATHENS RAW WATER
SCHEMATIC DESIGN

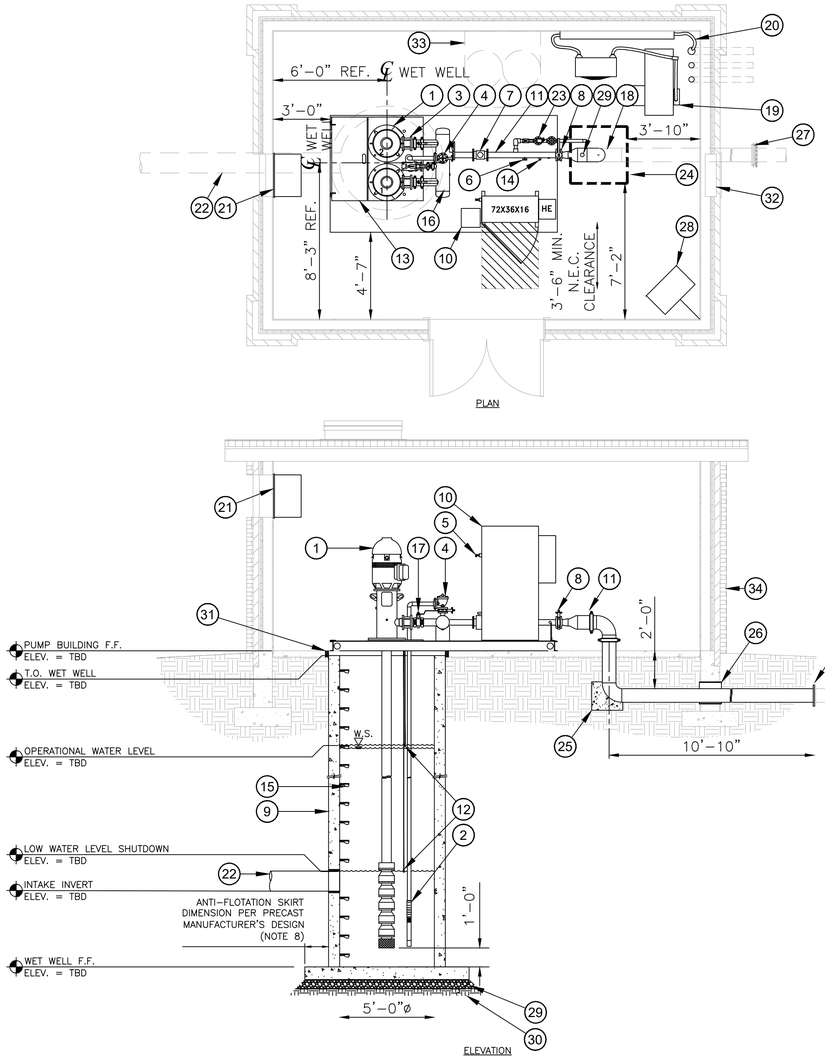
RAW WATER DETAILS

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PUMP STATION EQUIPMENT SCHEDULE

1. 40 HP MAIN PUMP AND MOTOR (2 TOTAL) EACH CAPABLE OF 450 GPM AT 100 PSI
2. 5 HP SUBMERSIBLE PRESSURE MAINTENANCE PUMP AND MOTOR
3. WAFER TYPE CHECK VALVE (3 TOTAL) SIZED TO MATCH PIPE
4. PRESSURE RELIEF VALVE WITH DISCHARGE PIPED TO WET WELL
5. PRESSURE TRANSDUCER (CABINET MOUNTED)
6. PRESSURE GAUGE 2.5" 150 PSI LIQUID FILLED
7. ELECTROMAGNETIC FLOW METER. PROVIDE A MINIMUM STRAIGHT RUN OF PIPE 5X THE PIPE DIAMETER UPSTREAM AND 3X THE PIPE DIAMETER DOWNSTREAM OF THE FLOW METER TO ACHIEVE THE PROPER FLOW REGIME
8. STATION ISOLATION BUTTERFLY VALVE
9. 5' DIAMETER PRECAST CONCRETE WET WELL WITH LADDER RUNGS AT ACCESS HATCH
10. ELECTRICAL STARTER, TIMERS, AND CONTROL CABINET (AIR-CONDITIONED)
11. PAINTED STEEL DISCHARGE PIPE AND FITTINGS
12. WATER LEVEL SENSOR PROBE
13. WET WELL HATCH
14. FERTIGATION PORT 1/2" THREAD-O-LET (T.O.L) PLUGGED
15. LADDER RUNGS, AMERICAN STEP COMPANY ML-13-NCR
16. DISCHARGE MANIFOLD
17. PUMP DISCHARGE ISOLATION VALVE (1 OF 3)
18. FLANGED STEEL DROP PIPE WITH SWIVEL VICTAULIC CONNECTION
19. AERATION COMPRESSOR AND EQUIPMENT
20. AERATION TUBES TO LAKE DIFFUSERS. COORDINATE INSTALLATION OF CONDUIT WITH BUILDING
21. THERMOSTATICALLY CONTROLLED EXHAUST FAN WITH WALL COLLAR AND DAMPER, MAKE/MODEL TBD, XXXX CFM, XXX VAC
22. INTAKE PIPE FROM INTAKE SCREEN ASSEMBLY
23. 2" SCH 40 STEEL INTAKE SCREEN FLUSH PIPE, WYE STRAINER, SOLENOID VALVE, DIRECT ACTING PRESSURE REDUCING VALVE, AND THREADED STEEL REDUCER. MAKE CONNECTION TO INTAKE SCREEN ASSEMBLY IN POND. TRANSITION TO POLYETHYLENE PIPE.
24. 3' X 3' CONCRETE CAST OUT FOR PLACEMENT OF DROP PIPE. COORDINATE FOR INSTALLATION OF DROP PIPE PRIOR TO INSTALLATION OF SLAB. ADJUST PLACEMENT OF PIPES TO ACCOMMODATE CONNECTION TO PUMP SKID AND FILL VOID WITH CONCRETE FLOW FILL. COORDINATE WITH ARCHITECTURAL PLANS.
25. 3'X3'X2' CONCRETE THRUST BLOCK
26. XX" SCH 40 STEEL PIPE SLEEVE THROUGH FOUNDATION, LENGTH AS REQUIRED
27. XX" FLANGED END CONNECTION, REFER TO IRRIGATION MAINLINE DRAWINGS.
28. 5 kW STATION HEATER, WALL MOUNTED, THERMOSTATICALLY CONTROLLED
29. WINTERIZATION CONNECTION: 2-INCH THREAD-O-LET WITH BRONZE BALL VALVE AND THREADED STEEL NIPPLE AND CAP
29. 6-INCH DEPTH OF ¾" GRAVEL
30. UNEXCAVATED SUBGRADE OR 95% SPD COMPACTED SUBGRADE
31. 1-1/2-INCH THICK EXPANSION JOINT
32. INTAKE LOUVER, TBD
33. SPACE FOR FUTURE TEA COMPOST SYSTEM
34. PUMP BUILDING, TBD

1 PUMP STATION DETAIL

N.T.S.

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UNIVERSITY OF COLORADO NEWTON AND ATHENS RAW WATER

SCHEMATIC DESIGN

RAW WATER DETAILS

PRELIMINARY DESIGN NOT FOR CONSTRUCTION

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